Exploratory Data Analysis

Registration Number: 2201583

Loading Libraries

```
library(ggplot2)
library(plotly)
```

```
##
## Attaching package: 'plotly'
```

```
## The following object is masked from 'package:ggplot2':
##
##     last_plot
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

```
## The following object is masked from 'package:graphics':
##
##     layout
```

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
## ✔ dplyr     1.1.2     ✔ readr     2.1.4
## ✔ forcats   1.0.0     ✔ stringr   1.5.0
## ✔ lubridate 1.9.2     ✔ tibble    3.2.1
## ✔ purrr     1.0.1     ✔ tidyr     1.3.0
```

```
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
## ✖ dplyr::filter() masks plotly::filter(), stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all con
flicts to become errors
```

```
library(leaflet)
library(magrittr)
```

```
##
## Attaching package: 'magrittr'
##
## The following object is masked from 'package:purrr':
##
##     set_names
##
## The following object is masked from 'package:tidyr':
##
##     extract
```

```
library(reshape2)
```

```
##
## Attaching package: 'reshape2'
##
## The following object is masked from 'package:tidyr':
##
##     smiths
```

```
library(leaflet)
```

Loading the Dataset

```
#First we are loading our dataset
data <- read.csv("F://data_exploration//dataset//police_Dataset.csv")
head(data)
```

```
##   INCIDENT_DATE INCIDENT_TIME    UOF_NUMBER OFFICER_ID OFFICER_GENDER
## 1    OCCURRED_D    OCCURRED_T        UOFNum CURRENT_BA         OffSex
## 2        9/3/16    4:14:00 AM         37702      10810           Male
## 3       3/22/16   11:00:00 PM         33413       7706           Male
## 4       5/22/16    1:29:00 PM         34567      11014           Male
## 5       1/10/16    8:55:00 PM         31460       6692           Male
## 6       11/8/16    2:30:00 AM  37879, 37898       9844           Male
##   OFFICER_RACE OFFICER_HIRE_DATE OFFICER_YEARS_ON_FORCE OFFICER_INJURY
## 1      OffRace           HIRE_DT    INCIDENT_DATE_LESS_     OFF_INJURE
## 2        Black            5/7/14                      2             No
## 3        White            1/8/99                     17            Yes
## 4        Black           5/20/15                      1             No
## 5        Black           7/29/91                     24             No
## 6        White           10/4/09                      7             No
##            OFFICER_INJURY_TYPE OFFICER_HOSPITALIZATION SUBJECT_ID SUBJECT_RACE
## 1               OFF_INJURE_DESC              OFF_HOSPIT     CitNum      CitRace
## 2 No injuries noted or visible                      No      46424        Black
## 3                 Sprain/Strain                     Yes      44324     Hispanic
## 4 No injuries noted or visible                      No      45126     Hispanic
## 5 No injuries noted or visible                      No      43150     Hispanic
## 6 No injuries noted or visible                      No      47307        Black
##   SUBJECT_GENDER SUBJECT_INJURY          SUBJECT_INJURY_TYPE
## 1         CitSex     CIT_INJURE             SUBJ_INJURE_DESC
## 2         Female            Yes      Non-Visible Injury/Pain
## 3           Male             No No injuries noted or visible
## 4           Male             No No injuries noted or visible
## 5           Male            Yes               Laceration/Cut
## 6           Male             No No injuries noted or visible
##   SUBJECT_WAS_ARRESTED SUBJECT_DESCRIPTION          SUBJECT_OFFENSE
## 1           CIT_ARREST          CIT_INFL_A               CitChargeT
## 2                  Yes   Mentally unstable                    APOWW
## 3                  Yes   Mentally unstable                    APOWW
## 4                  Yes             Unknown                    APOWW
## 5                  Yes FD-Unknown if Armed           Evading Arrest
## 6                  Yes             Unknown Other Misdemeanor Arrest
##   REPORTING_AREA BEAT SECTOR      DIVISION LOCATION_DISTRICT STREET_NUMBER
## 1             RA BEAT SECTOR      DIVISION         DIST_NAME      STREET_N
## 2           2062  134    130       CENTRAL               D14           211
## 3           1197  237    230     NORTHEAST                D9          7647
## 4           4153  432    430     SOUTHWEST                D6           716
## 5           4523  641    640 NORTH CENTRAL               D11          5600
## 6           2167  346    340     SOUTHEAST                D7          4600
##    STREET_NAME STREET_DIRECTION STREET_TYPE
## 1       STREET         street_g    street_t
## 2        Ervay                N         St.
## 3     Ferguson             NULL         Rd.
## 4 bimebella dr             NULL         Ln.
## 5          LBJ             NULL       Frwy.
## 6    Malcolm X                S       Blvd.
##   LOCATION_FULL_STREET_ADDRESS_OR_INTERSECTION LOCATION_CITY LOCATION_STATE
## 1                               Street Address          City          State
## 2                                 211 N ERVAY ST        Dallas             TX
## 3                              7647 FERGUSON RD        Dallas             TX
## 4                              716 BIMEBELLA LN        Dallas             TX
## 5                                5600 L B J FWY        Dallas             TX
## 6                         4600 S MALCOLM X BLVD        Dallas             TX
```

```
##   LOCATION_LATITUDE LOCATION_LONGITUDE INCIDENT_REASON REASON_FOR_FORCE
## 1          Latitude          Longitude      SERVICE_TY       UOF_REASON
## 2         32.782205         -96.797461          Arrest           Arrest
## 3         32.798978         -96.717493          Arrest           Arrest
## 4          32.73971          -96.92519          Arrest           Arrest
## 5                                               Arrest           Arrest
## 6                                               Arrest           Arrest
##     TYPE_OF_FORCE_USED1 TYPE_OF_FORCE_USED2 TYPE_OF_FORCE_USED3
## 1            ForceType1          ForceType2          ForceType3
## 2 Hand/Arm/Elbow Strike
## 3           Joint Locks
## 4     Take Down - Group
## 5        K-9 Deployment
## 6        Verbal Command     Take Down - Arm
##   TYPE_OF_FORCE_USED4 TYPE_OF_FORCE_USED5 TYPE_OF_FORCE_USED6
## 1          ForceType4          ForceType5          ForceType6
## 2
## 3
## 4
## 5
## 6
##   TYPE_OF_FORCE_USED7 TYPE_OF_FORCE_USED8 TYPE_OF_FORCE_USED9
## 1          ForceType7          ForceType8          ForceType9
## 2
## 3
## 4
## 5
## 6
##   TYPE_OF_FORCE_USED10 NUMBER_EC_CYCLES FORCE_EFFECTIVE
## 1          ForceType10       Cycles_Num      ForceEffec
## 2                                  NULL             Yes
## 3                                  NULL             Yes
## 4                                  NULL             Yes
## 5                                  NULL             Yes
## 6                                  NULL         No, Yes
```

Data Preprocessing

We will remove the first row of dataset because it is repeating the main headings.

```
#removing first row
data <- data[-1, ]
head(data)
```

```
##   INCIDENT_DATE INCIDENT_TIME    UOF_NUMBER OFFICER_ID OFFICER_GENDER
## 2        9/3/16    4:14:00 AM         37702      10810           Male
## 3       3/22/16   11:00:00 PM         33413       7706           Male
## 4       5/22/16    1:29:00 PM         34567      11014           Male
## 5       1/10/16    8:55:00 PM         31460       6692           Male
## 6       11/8/16    2:30:00 AM  37879, 37898       9844           Male
## 7       9/11/16    7:20:00 PM         36724       9855           Male
##   OFFICER_RACE OFFICER_HIRE_DATE OFFICER_YEARS_ON_FORCE OFFICER_INJURY
## 2        Black            5/7/14                      2             No
## 3        White            1/8/99                     17            Yes
## 4        Black           5/20/15                      1             No
## 5        Black           7/29/91                     24             No
## 6        White           10/4/09                      7             No
## 7        White           6/10/09                      7             No
##            OFFICER_INJURY_TYPE OFFICER_HOSPITALIZATION SUBJECT_ID SUBJECT_RACE
## 2 No injuries noted or visible                      No      46424        Black
## 3               Sprain/Strain                      Yes      44324     Hispanic
## 4 No injuries noted or visible                      No      45126     Hispanic
## 5 No injuries noted or visible                      No      43150     Hispanic
## 6 No injuries noted or visible                      No      47307        Black
## 7 No injuries noted or visible                      No      46549        White
##   SUBJECT_GENDER SUBJECT_INJURY          SUBJECT_INJURY_TYPE
## 2         Female            Yes      Non-Visible Injury/Pain
## 3           Male             No No injuries noted or visible
## 4           Male             No No injuries noted or visible
## 5           Male            Yes               Laceration/Cut
## 6           Male             No No injuries noted or visible
## 7         Female             No No injuries noted or visible
##   SUBJECT_WAS_ARRESTED SUBJECT_DESCRIPTION          SUBJECT_OFFENSE
## 2                  Yes   Mentally unstable                    APOWW
## 3                  Yes   Mentally unstable                    APOWW
## 4                  Yes             Unknown                    APOWW
## 5                  Yes FD-Unknown if Armed           Evading Arrest
## 6                  Yes             Unknown Other Misdemeanor Arrest
## 7                  Yes             Unknown               Assault/FV
##   REPORTING_AREA BEAT SECTOR      DIVISION LOCATION_DISTRICT STREET_NUMBER
## 2           2062  134    130       CENTRAL               D14           211
## 3           1197  237    230     NORTHEAST                D9          7647
## 4           4153  432    430     SOUTHWEST                D6           716
## 5           4523  641    640 NORTH CENTRAL               D11          5600
## 6           2167  346    340     SOUTHEAST                D7          4600
## 7           1134  235    230     NORTHEAST                D9          1234
##    STREET_NAME STREET_DIRECTION STREET_TYPE
## 2        Ervay                N         St.
## 3     Ferguson             NULL         Rd.
## 4 bimebella dr             NULL         Ln.
## 5          LBJ             NULL       Frwy.
## 6    Malcolm X                S       Blvd.
## 7        Peavy             NULL         Rd.
##   LOCATION_FULL_STREET_ADDRESS_OR_INTERSECTION LOCATION_CITY LOCATION_STATE
## 2                                 211 N ERVAY ST        Dallas             TX
## 3                               7647 FERGUSON RD        Dallas             TX
## 4                               716 BIMEBELLA LN        Dallas             TX
## 5                                 5600 L B J FWY        Dallas             TX
## 6                          4600 S MALCOLM X BLVD        Dallas             TX
## 7                                  1234 PEAVY RD        Dallas             TX
```

```
##   LOCATION_LATITUDE LOCATION_LONGITUDE INCIDENT_REASON REASON_FOR_FORCE
## 2         32.782205         -96.797461          Arrest           Arrest
## 3         32.798978         -96.717493          Arrest           Arrest
## 4          32.73971          -96.92519          Arrest           Arrest
## 5                                               Arrest           Arrest
## 6                                               Arrest           Arrest
## 7         32.837527         -96.695566          Arrest           Arrest
##      TYPE_OF_FORCE_USED1 TYPE_OF_FORCE_USED2 TYPE_OF_FORCE_USED3
## 2  Hand/Arm/Elbow Strike
## 3            Joint Locks
## 4      Take Down - Group
## 5         K-9 Deployment
## 6         Verbal Command     Take Down - Arm
## 7 Hand Controlled Escort
##   TYPE_OF_FORCE_USED4 TYPE_OF_FORCE_USED5 TYPE_OF_FORCE_USED6
## 2
## 3
## 4
## 5
## 6
## 7
##   TYPE_OF_FORCE_USED7 TYPE_OF_FORCE_USED8 TYPE_OF_FORCE_USED9
## 2
## 3
## 4
## 5
## 6
## 7
##   TYPE_OF_FORCE_USED10 NUMBER_EC_CYCLES FORCE_EFFECTIVE
## 2                                  NULL             Yes
## 3                                  NULL             Yes
## 4                                  NULL             Yes
## 5                                  NULL             Yes
## 6                                  NULL         No, Yes
## 7                                  NULL             Yes
```

```
#checking the dimensions of the data how many rows and columns are present.
dim(data)
```

```
## [1] 2383   47
```

We have total 2383 rows and 47 columns

```
#lets check the datatypes ofcolumns by using str function
str(data)
```

```
## 'data.frame':	2383 obs. of  47 variables:
##  $ INCIDENT_DATE                                : chr  "9/3/16" "3/22/16" "5/2
2/16" "1/10/16" ...
##  $ INCIDENT_TIME                                : chr  "4:14:00 AM" "11:00:00 P
M" "1:29:00 PM" "8:55:00 PM" ...
##  $ UOF_NUMBER                                   : chr  "37702" "33413" "34567"
"31460" ...
##  $ OFFICER_ID                                   : chr  "10810" "7706" "11014"
"6692" ...
##  $ OFFICER_GENDER                               : chr  "Male" "Male" "Male" "Ma
le" ...
##  $ OFFICER_RACE                                 : chr  "Black" "White" "Black"
"Black" ...
##  $ OFFICER_HIRE_DATE                            : chr  "5/7/14" "1/8/99" "5/20/
15" "7/29/91" ...
##  $ OFFICER_YEARS_ON_FORCE                       : chr  "2" "17" "1" "24" ...
##  $ OFFICER_INJURY                               : chr  "No" "Yes" "No" "No" ...
##  $ OFFICER_INJURY_TYPE                          : chr  "No injuries noted or vi
sible" "Sprain/Strain" "No injuries noted or visible" "No injuries noted or visib
le" ...
##  $ OFFICER_HOSPITALIZATION                      : chr  "No" "Yes" "No" "No" ...
##  $ SUBJECT_ID                                   : chr  "46424" "44324" "45126"
"43150" ...
##  $ SUBJECT_RACE                                 : chr  "Black" "Hispanic" "Hisp
anic" "Hispanic" ...
##  $ SUBJECT_GENDER                               : chr  "Female" "Male" "Male"
"Male" ...
##  $ SUBJECT_INJURY                               : chr  "Yes" "No" "No" "Yes"
...
##  $ SUBJECT_INJURY_TYPE                          : chr  "Non-Visible Injury/Pai
n" "No injuries noted or visible" "No injuries noted or visible" "Laceration/Cut"
...
##  $ SUBJECT_WAS_ARRESTED                         : chr  "Yes" "Yes" "Yes" "Yes"
...
##  $ SUBJECT_DESCRIPTION                          : chr  "Mentally unstable" "Men
tally unstable" "Unknown" "FD-Unknown if Armed" ...
##  $ SUBJECT_OFFENSE                              : chr  "APOWW" "APOWW" "APOWW"
"Evading Arrest" ...
##  $ REPORTING_AREA                               : chr  "2062" "1197" "4153" "45
23" ...
##  $ BEAT                                         : chr  "134" "237" "432" "641"
...
##  $ SECTOR                                       : chr  "130" "230" "430" "640"
...
##  $ DIVISION                                     : chr  "CENTRAL" "NORTHEAST" "S
OUTHWEST" "NORTH CENTRAL" ...
##  $ LOCATION_DISTRICT                            : chr  "D14" "D9" "D6" "D11"
...
##  $ STREET_NUMBER                                : chr  "211" "7647" "716" "560
0" ...
##  $ STREET_NAME                                  : chr  "Ervay" "Ferguson" "bime
bella dr" "LBJ" ...
##  $ STREET_DIRECTION                             : chr  "N" "NULL" "NULL" "NULL"
...
##  $ STREET_TYPE                                  : chr  "St." "Rd." "Ln." "Frw
y." ...
```

```
##  $ LOCATION_FULL_STREET_ADDRESS_OR_INTERSECTION: chr  "211 N ERVAY ST" "7647 F
ERGUSON RD" "716 BIMEBELLA LN" "5600 L B J FWY" ...
##  $ LOCATION_CITY                               : chr  "Dallas" "Dallas" "Dalla
s" "Dallas" ...
##  $ LOCATION_STATE                              : chr  "TX" "TX" "TX" "TX" ...
##  $ LOCATION_LATITUDE                           : chr  "32.782205" "32.798978"
"32.73971" "" ...
##  $ LOCATION_LONGITUDE                          : chr  "-96.797461" "-96.71749
3" "-96.92519" "" ...
##  $ INCIDENT_REASON                             : chr  "Arrest" "Arrest" "Arres
t" "Arrest" ...
##  $ REASON_FOR_FORCE                            : chr  "Arrest" "Arrest" "Arres
t" "Arrest" ...
##  $ TYPE_OF_FORCE_USED1                         : chr  "Hand/Arm/Elbow Strike"
"Joint Locks" "Take Down - Group" "K-9 Deployment" ...
##  $ TYPE_OF_FORCE_USED2                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED3                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED4                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED5                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED6                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED7                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED8                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED9                         : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED10                        : chr  "" "" "" "" ...
##  $ NUMBER_EC_CYCLES                            : chr  "NULL" "NULL" "NULL" "NU
LL" ...
##  $ FORCE_EFFECTIVE                             : chr  " Yes" " Yes" " Yes" " Y
es" ...
```

The all columns are on char datatype we need to change their datatypes.before that its a time series dataset so we need to find is there any missing values in date and time or not.

```
Itime_na <- sum(data$INCIDENT_TIME == "NULL", na.rm = TRUE)
cat("Number of Null values in INCIDENT_TIME are: ",Itime_na)
```

```
## Number of Null values in INCIDENT_TIME are:  10
```

```
Idate_na <- sum(data$INCIDENT_DATE == "NULL", na.rm = TRUE)
cat("Number of Null values in INCIDENT_DATE are: ",Idate_na)
```

```
## Number of Null values in INCIDENT_DATE are:  0
```

There is 10 Null values in INCIDENT_TIME column so we use complete.cases function to identify rows with complete data.

```
data$INCIDENT_TIME <- ifelse(data$INCIDENT_TIME == "NULL", NA, data$INCIDENT_TIM
E)
data <- data[complete.cases(data$INCIDENT_TIME),]

Itime_na <- sum(data$INCIDENT_TIME == "NULL", na.rm = TRUE)
cat("Number of Null values in INCIDENT_TIME are: ",Itime_na)
```

```
## Number of Null values in INCIDENT_TIME are:  0
```

Converting Datatypes

```
#lets convert the data type of date and time and combining 2 columns date and tim
e and make it as a one column.
data$INCIDENT_DATETIME <- as.POSIXct(paste(data$INCIDENT_DATE, data$INCIDENT_TIM
E),
                                     format="%m/%d/%Y %I:%M:%S %p", tz="UTC")
data$INCIDENT_DATETIME <- format(data$INCIDENT_DATETIME, "%y-%m-%d %H:%M:%S")
data <- data %>% select(-c(INCIDENT_DATE, INCIDENT_TIME))
data$INCIDENT_DATETIME <- as.POSIXct(data$INCIDENT_DATETIME, format = "%y-%m-%d %
H:%M:%S")



# converting Officer hire date column into date format
data$OFFICER_HIRE_DATE <- as.Date(data$OFFICER_HIRE_DATE, format="%m/%d/%Y")
data$OFFICER_HIRE_DATE <- format(data$OFFICER_HIRE_DATE, "%y-%m-%d")
data$OFFICER_HIRE_DATE <- as.POSIXct(data$OFFICER_HIRE_DATE, format = "%y-%m-%d")
```

```
#Now we are converting all the columns which has to be int data type
data$OFFICER_ID <- as.integer(data$OFFICER_ID)
data$OFFICER_YEARS_ON_FORCE <- as.integer(data$OFFICER_YEARS_ON_FORCE)
data$SUBJECT_ID <- as.integer(data$SUBJECT_ID)
data$REPORTING_AREA <- as.integer(data$REPORTING_AREA)
data$BEAT <- as.integer(data$BEAT)
data$SECTOR <- as.integer(data$SECTOR)
data$STREET_NUMBER <- as.integer(data$STREET_NUMBER)
```

```
#Now we are converting all the columns which has to be float datatype like longit
ude and latitude
data$LOCATION_LATITUDE <- as.numeric(data$LOCATION_LATITUDE)
data$LOCATION_LONGITUDE <- as.numeric(data$LOCATION_LONGITUDE)
```

Lets check all the data types again

```
str(data)
```

```
## 'data.frame':	2373 obs. of  46 variables:
##  $ UOF_NUMBER                                 : chr  "37702" "33413" "34567"
"31460" ...
##  $ OFFICER_ID                                 : int  10810 7706 11014 6692 98
44 9855 9881 9058 10381 9705 ...
##  $ OFFICER_GENDER                             : chr  "Male" "Male" "Male" "Ma
le" ...
##  $ OFFICER_RACE                               : chr  "Black" "White" "Black"
"Black" ...
##  $ OFFICER_HIRE_DATE                          : POSIXct, format: "2014-05-07"
"1999-01-08" ...
##  $ OFFICER_YEARS_ON_FORCE                     : int  2 17 1 24 7 7 7 9 4 8
...
##  $ OFFICER_INJURY                             : chr  "No" "Yes" "No" "No" ...
##  $ OFFICER_INJURY_TYPE                        : chr  "No injuries noted or vi
sible" "Sprain/Strain" "No injuries noted or visible" "No injuries noted or visib
le" ...
##  $ OFFICER_HOSPITALIZATION                    : chr  "No" "Yes" "No" "No" ...
##  $ SUBJECT_ID                                 : int  46424 44324 45126 43150
47307 46549 47555 44172 43723 47889 ...
##  $ SUBJECT_RACE                               : chr  "Black" "Hispanic" "Hisp
anic" "Hispanic" ...
##  $ SUBJECT_GENDER                             : chr  "Female" "Male" "Male"
"Male" ...
##  $ SUBJECT_INJURY                             : chr  "Yes" "No" "No" "Yes"
...
##  $ SUBJECT_INJURY_TYPE                        : chr  "Non-Visible Injury/Pai
n" "No injuries noted or visible" "No injuries noted or visible" "Laceration/Cut"
...
##  $ SUBJECT_WAS_ARRESTED                       : chr  "Yes" "Yes" "Yes" "Yes"
...
##  $ SUBJECT_DESCRIPTION                        : chr  "Mentally unstable" "Men
tally unstable" "Unknown" "FD-Unknown if Armed" ...
##  $ SUBJECT_OFFENSE                            : chr  "APOWW" "APOWW" "APOWW"
"Evading Arrest" ...
##  $ REPORTING_AREA                             : int  2062 1197 4153 4523 2167
1134 2049 3122 2072 4403 ...
##  $ BEAT                                       : int  134 237 432 641 346 235
132 515 133 614 ...
##  $ SECTOR                                     : int  130 230 430 640 340 230
130 510 130 610 ...
##  $ DIVISION                                   : chr  "CENTRAL" "NORTHEAST" "S
OUTHWEST" "NORTH CENTRAL" ...
##  $ LOCATION_DISTRICT                          : chr  "D14" "D9" "D6" "D11"
...
##  $ STREET_NUMBER                              : int  211 7647 716 5600 4600 1
234 511 4709 300 18600 ...
##  $ STREET_NAME                                : chr  "Ervay" "Ferguson" "bime
bella dr" "LBJ" ...
##  $ STREET_DIRECTION                           : chr  "N" "NULL" "NULL" "NULL"
...
##  $ STREET_TYPE                                : chr  "St." "Rd." "Ln." "Frw
y." ...
##  $ LOCATION_FULL_STREET_ADDRESS_OR_INTERSECTION: chr  "211 N ERVAY ST" "7647 F
ERGUSON RD" "716 BIMEBELLA LN" "5600 L B J FWY" ...
##  $ LOCATION_CITY                              : chr  "Dallas" "Dallas" "Dalla
```

```
s" "Dallas" ...
##  $ LOCATION_STATE                           : chr  "TX" "TX" "TX" "TX" ...
##  $ LOCATION_LATITUDE                        : num  32.8 32.8 32.7 NA NA ...
##  $ LOCATION_LONGITUDE                       : num  -96.8 -96.7 -96.9 NA NA
...
##  $ INCIDENT_REASON                          : chr  "Arrest" "Arrest" "Arres
t" "Arrest" ...
##  $ REASON_FOR_FORCE                         : chr  "Arrest" "Arrest" "Arres
t" "Arrest" ...
##  $ TYPE_OF_FORCE_USED1                      : chr  "Hand/Arm/Elbow Strike"
"Joint Locks" "Take Down - Group" "K-9 Deployment" ...
##  $ TYPE_OF_FORCE_USED2                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED3                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED4                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED5                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED6                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED7                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED8                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED9                      : chr  "" "" "" "" ...
##  $ TYPE_OF_FORCE_USED10                     : chr  "" "" "" "" ...
##  $ NUMBER_EC_CYCLES                         : chr  "NULL" "NULL" "NULL" "NU
LL" ...
##  $ FORCE_EFFECTIVE                          : chr  " Yes" " Yes" " Yes" " Y
es" ...
##  $ INCIDENT_DATETIME                        : POSIXct, format: "2016-09-03
04:14:00" "2016-03-22 23:00:00" ...
```

All datatypes are converted lets find the summary of the dataset

```
#Finding the descriptive statistics of the data by using summary function
summary(data)
```

```
##   UOF_NUMBER          OFFICER_ID    OFFICER_GENDER     OFFICER_RACE      
##  Length:2373        Min.   :    0   Length:2373        Length:2373       
##  Class :character   1st Qu.: 8908   Class :character   Class :character  
##  Mode  :character   Median :10115   Mode  :character   Mode  :character  
##                     Mean   : 9574                                        
##                     3rd Qu.:10710                                        
##                     Max.   :11170                                        
##                                                                          
##  OFFICER_HIRE_DATE                OFFICER_YEARS_ON_FORCE OFFICER_INJURY    
##  Min.   :1979-08-20 00:00:00.00   Min.   : 0.000         Length:2373       
##  1st Qu.:2006-08-11 00:00:00.00   1st Qu.: 3.000         Class :character  
##  Median :2010-06-09 00:00:00.00   Median : 6.000         Mode  :character  
##  Mean   :2008-05-29 15:48:00.91   Mean   : 8.038                           
##  3rd Qu.:2013-11-06 00:00:00.00   3rd Qu.:10.000                           
##  Max.   :2016-04-06 00:00:00.00   Max.   :36.000                           
##                                                                            
##  OFFICER_INJURY_TYPE OFFICER_HOSPITALIZATION   SUBJECT_ID    SUBJECT_RACE      
##  Length:2373         Length:2373             Min.   :    0   Length:2373       
##  Class :character    Class :character        1st Qu.:43312   Class :character  
##  Mode  :character    Mode  :character        Median :44575   Mode  :character  
##                                              Mean   :40270                     
##                                              3rd Qu.:46092                     
##                                              Max.   :47972                     
##                                                                                
##  SUBJECT_GENDER     SUBJECT_INJURY     SUBJECT_INJURY_TYPE SUBJECT_WAS_ARRESTE
D
##  Length:2373        Length:2373        Length:2373         Length:2373        
##  Class :character   Class :character   Class :character    Class :character   
##  Mode  :character   Mode  :character   Mode  :character    Mode  :character   
##                                                                               
##                                                                               
##                                                                               
##                                                                               
##  SUBJECT_DESCRIPTION SUBJECT_OFFENSE    REPORTING_AREA      BEAT      
##  Length:2373         Length:2373        Min.   :1001   Min.   :111.0  
##  Class :character    Class :character   1st Qu.:2015   1st Qu.:216.0  
##  Mode  :character    Mode  :character   Median :2403   Median :351.0  
##                                         Mean   :3195   Mean   :393.2  
##                                         3rd Qu.:4327   3rd Qu.:612.0  
##                                         Max.   :9611   Max.   :757.0  
##                                                                       
##      SECTOR        DIVISION         LOCATION_DISTRICT  STREET_NUMBER  
##  Min.   :110.0   Length:2373        Length:2373        Min.   :    0  
##  1st Qu.:210.0   Class :character   Class :character   1st Qu.: 1700  
##  Median :350.0   Mode  :character   Mode  :character   Median : 3400  
##  Mean   :389.4                                         Mean   : 4898  
##  3rd Qu.:610.0                                         3rd Qu.: 7532  
##  Max.   :750.0                                         Max.   :54023  
##                                                                       
##  STREET_NAME        STREET_DIRECTION   STREET_TYPE       
##  Length:2373        Length:2373        Length:2373       
##  Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character  
##                                                          
##                                                          
##                                                          
```

```
##
## LOCATION_FULL_STREET_ADDRESS_OR_INTERSECTION LOCATION_CITY
## Length:2373                                  Length:2373
## Class :character                             Class :character
## Mode  :character                             Mode  :character
##
##
##
##
## LOCATION_STATE     LOCATION_LATITUDE LOCATION_LONGITUDE INCIDENT_REASON
## Length:2373        Min.   :32.63     Min.   :-96.96     Length:2373
## Class :character   1st Qu.:32.74     1st Qu.:-96.82     Class :character
## Mode  :character   Median :32.78     Median :-96.79     Mode  :character
##                    Mean   :32.80     Mean   :-96.78
##                    3rd Qu.:32.86     3rd Qu.:-96.75
##                    Max.   :33.02     Max.   :-96.57
##                    NA's   :55        NA's   :55
## REASON_FOR_FORCE   TYPE_OF_FORCE_USED1 TYPE_OF_FORCE_USED2 TYPE_OF_FORCE_USED
3
## Length:2373        Length:2373         Length:2373         Length:2373
## Class :character   Class :character    Class :character    Class :character
## Mode  :character   Mode  :character    Mode  :character    Mode  :character
##
##
##
##
## TYPE_OF_FORCE_USED4 TYPE_OF_FORCE_USED5 TYPE_OF_FORCE_USED6
## Length:2373         Length:2373         Length:2373
## Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character
##
##
##
##
## TYPE_OF_FORCE_USED7 TYPE_OF_FORCE_USED8 TYPE_OF_FORCE_USED9
## Length:2373         Length:2373         Length:2373
## Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character
##
##
##
##
## TYPE_OF_FORCE_USED10 NUMBER_EC_CYCLES   FORCE_EFFECTIVE
## Length:2373          Length:2373        Length:2373
## Class :character     Class :character   Class :character
## Mode  :character     Mode  :character   Mode  :character
##
##
##
##
## INCIDENT_DATETIME
## Min.   :2016-01-01 00:11:00.00
## 1st Qu.:2016-03-11 23:30:00.00
## Median :2016-05-31 03:00:00.00
## Mean   :2016-06-10 23:07:36.10
## 3rd Qu.:2016-09-06 11:20:00.00
```

```
## Max. :2016-12-31 23:19:00.00
##
```

Creating one month column with this months value we can do some analysis on montly basis.

```
#we create a month column
month <- as.numeric(format(data$INCIDENT_DATETIME,'%m'))
head(month)
```

```
## [1] 9 3 5 1 11 9
```

```
#combining this month column with main dataset
data <- cbind(data,month)
```

Now we remove some columns which are having mostly null values which are not relevant for our analysis.

```
data <- subset(data, select = -c(TYPE_OF_FORCE_USED1,TYPE_OF_FORCE_USED2,TYPE_OF_
FORCE_USED3,TYPE_OF_FORCE_USED4,TYPE_OF_FORCE_USED5,
                                 TYPE_OF_FORCE_USED6,TYPE_OF_FORCE_USED7,TYPE_OF_
FORCE_USED8,TYPE_OF_FORCE_USED9,TYPE_OF_FORCE_USED10,NUMBER_EC_CYCLES,
                                 FORCE_EFFECTIVE,STREET_DIRECTION))
```

Now we fill na values

```
data <- na.omit(data)
```

Data Analysis

lets find how many categorical features we have and how many unique values they have.

```
# finding how many categories we have in a categorical columns
cat_cols <- c("OFFICER_GENDER", "OFFICER_RACE", "OFFICER_INJURY", "OFFICER_INJURY
_TYPE",
              "OFFICER_HOSPITALIZATION", "SUBJECT_RACE", "SUBJECT_GENDER", "SUBJE
CT_INJURY",
              "SUBJECT_INJURY_TYPE", "SUBJECT_WAS_ARRESTED", "SUBJECT_DESCRIPTIO
N", "SUBJECT_OFFENSE",
              "DIVISION", "LOCATION_DISTRICT", "STREET_NAME", "STREET_TYPE", "INC
IDENT_REASON",
              "REASON_FOR_FORCE", "TYPE_OF_FORCE_USED1", "FORCE_EFFECTIVE", "NUMB
ER_EC_CYCLES")

for (col in cat_cols) {
  cat_count <- length(unique(data[[col]]))
  cat_info <- paste(col, ": ", cat_count, " unique categories")
  print(cat_info)
}
```

```
## [1] "OFFICER_GENDER :  2  unique categories"
## [1] "OFFICER_RACE :  6  unique categories"
## [1] "OFFICER_INJURY :  2  unique categories"
## [1] "OFFICER_INJURY_TYPE :  75  unique categories"
## [1] "OFFICER_HOSPITALIZATION :  2  unique categories"
## [1] "SUBJECT_RACE :  7  unique categories"
## [1] "SUBJECT_GENDER :  4  unique categories"
## [1] "SUBJECT_INJURY :  2  unique categories"
## [1] "SUBJECT_INJURY_TYPE :  189  unique categories"
## [1] "SUBJECT_WAS_ARRESTED :  2  unique categories"
## [1] "SUBJECT_DESCRIPTION :  15  unique categories"
## [1] "SUBJECT_OFFENSE :  538  unique categories"
## [1] "DIVISION :  7  unique categories"
## [1] "LOCATION_DISTRICT :  14  unique categories"
## [1] "STREET_NAME :  1046  unique categories"
## [1] "STREET_TYPE :  22  unique categories"
## [1] "INCIDENT_REASON :  14  unique categories"
## [1] "REASON_FOR_FORCE :  12  unique categories"
## [1] "TYPE_OF_FORCE_USED1 :  0  unique categories"
## [1] "FORCE_EFFECTIVE :  0  unique categories"
## [1] "NUMBER_EC_CYCLES :  0  unique categories"
```

Visualization

Lets create some plots and charts to better visualize and analyze the dataset we use ggplot2 to do that first finding how many cases we have in year 2016 using time series plot

```
#making the histogram to see the time series plot using datetime
my_hist <- plot_ly(data, x = ~INCIDENT_DATETIME, type = "histogram", nbinsx = 30)

my_hist <- layout(my_hist, xaxis = list(title = "Date"), yaxis = list(title = "Co
unt"), title = "Time Series Histogram")

my_hist
```

We can see that from 21st february to 6th march we have maximum cases of 144 from november to december we noticed that the cases are decreasing which means crime rate were going to be down after november. the start of 2017 was very good compare to 2016 there is a huge downfall of crimes can be seen in the graph.

lets analyze some data of criminals and police separately

Police officer data

```
#creating a plot to count the gender of officers
gender_count <- data.frame(table(data$OFFICER_GENDER))

my_plot <- plot_ly(gender_count, labels = ~Var1, values = ~Freq, type = "pie",
             marker = list(colors = c("#38C4D0", "#026F78"), line = list(color =
"#FFFFFF", width = 1))) %>%
  layout(title = "Count of Officers by Gender")

# Display the plot
my_plot
```

It can be seen that there are maximum Male officers which were on the case where as female officers are very less.around 89.8% of the officers are males and rest of 10.2% are females.

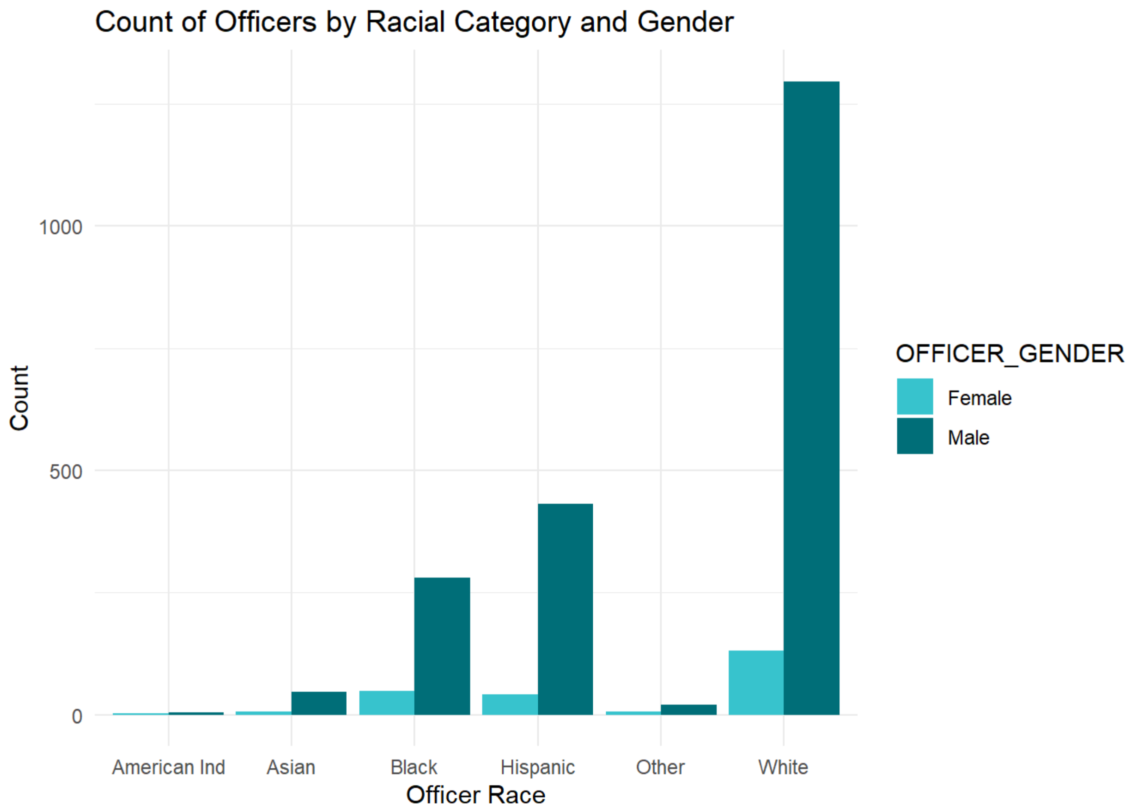
```
#creating a bar plot to see the count of officers by racial category
race_count <- data.frame(table(data$OFFICER_RACE))

my_plot <- plot_ly(race_count, x = ~Var1, y = ~Freq, type = "bar",
             marker = list(color = "#1f77b4", opacity = 0.7)) %>%
  layout(title = "Count of Officers by Racial categories",
         xaxis = list(title = "Race"),
         yaxis = list(title = "Count"))

# Display the plot
my_plot
```

Mostly officers belongs to the White racial category where as the American ind and asian were very less also wee need to see the gender of the officers for example: how many males officers are white and how many females

```
# craeting two table graph to see the genders with respect to racial category
race_gender_count <- data %>%
  group_by(OFFICER_RACE, OFFICER_GENDER) %>%
  summarize(n = n()) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'OFFICER_RACE'. You can override using the
## `.groups` argument.
```

```
ggplot(race_gender_count, aes(x = OFFICER_RACE, y = n, fill = OFFICER_GENDER)) +
  geom_bar(stat="identity", position="dodge") +
  xlab("Officer Race") +
  ylab("Count") +
  ggtitle("Count of Officers by Racial Category and Gender") +
  scale_fill_manual(values = c("#38C4D0", "#026F78")) +
  theme_minimal()
```



We can see that maximum males and females belongs to white racial category as their is the possibility that they must be the citizens of USA where as in Hispanic and Black categories contains the same amount of women officer.

In Every organization experience members are the pillars of that organization so on the same way lets check how many officers had how much experience.

```
# finding the experience of the officers
library(plotly)
library(ggplot2)

years_count <- data.frame(table(data$OFFICER_YEARS_ON_FORCE))

# using ggplot to create bar chart
ggplot(years_count, aes(x=Var1, y=Freq)) +
  geom_bar(stat="identity", fill="#38C4D0") +
  xlab("Years of Experience") +
  ylab("Count") +
  ggtitle("Count of Officers by Years of Experience") +
  geom_text(aes(label=Freq), vjust=1.6, color="black", size=3.5) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) -> p

my_plot <- ggplotly(p, tooltip = c("x", "y"))

# Customize the layout and hover info
my_plot %>%
  layout(title = "Count of Officers by Years of Experience",
         xaxis = list(title = "Years of Experience"),
         yaxis = list(title = "Count"),
         hovermode = "closest",
         hovertemplate = paste('%{x}: %{y}<br>Total Count: ', sum(years_count$Fre
q)))
```

```
## Warning: 'layout' objects don't have these attributes: 'hovertemplate'
## Valid attributes include:
## '_deprecated', 'activeshape', 'annotations', 'autosize', 'autotypenumbers', 'c
alendar', 'clickmode', 'coloraxis', 'colorscale', 'colorway', 'computed', 'datare
vision', 'dragmode', 'editrevision', 'editType', 'font', 'geo', 'grid', 'height',
'hidesources', 'hoverdistance', 'hoverlabel', 'hovermode', 'images', 'legend', 'm
apbox', 'margin', 'meta', 'metasrc', 'modebar', 'newshape', 'paper_bgcolor', 'plo
t_bgcolor', 'polar', 'scene', 'selectdirection', 'selectionrevision', 'separator
s', 'shapes', 'showlegend', 'sliders', 'smith', 'spikedistance', 'template', 'ter
nary', 'title', 'transition', 'uirevision', 'uniformtext', 'updatemenus', 'widt
h', 'xaxis', 'yaxis', 'barmode', 'bargap', 'mapType'
```

It can be seen that maximum police officers are fresh they have 2 to 3 years of experience. whereas the the people having more experience are very less in count.

Let see the count of officers work in each month

```
# finding the count of officers with respect to each month
officers_by_month <- data %>%
  group_by(month) %>%
  summarise(n = n_distinct(OFFICER_ID))

# using plotly to make interactive bar plot
plot_ly(officers_by_month, x = ~month, y = ~n, type = "bar",
        hovertemplate = "Month: %{x}<br>Officer Count: %{y}") %>%
  layout(title = "Count of Officers by Month",
         xaxis = list(title = "Month"),
         yaxis = list(title = "Officer Count"))
```

We can see that in the start of the year there were maximum police officers working but in the month of december there were less police officers who worked that might hapeen because of public holidays of cristmas.

lets find out which racial category wise officer experience.

```
#creating a box plot to see the officer experience with respect to racial categor
y
library(viridis)
```

```
## Loading required package: viridisLite
```

```
color_palette <- viridis(6, alpha = 0.7)

my_plot <- ggplot(data = data, aes(x = OFFICER_RACE, y = OFFICER_YEARS_ON_FORCE,
fill = OFFICER_RACE)) +
  geom_boxplot(color = "#757575", outlier.color = "#757575", alpha = 0.7) +
  scale_fill_manual(values = color_palette) +
  labs(x = "Officer Race", y = "Years on Force", title = "Years of Experience wit
h respect to Officer Race") +
  theme_minimal(base_size = 14) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        panel.background = element_blank(),
        axis.line = element_line(color = "#757575"),
        legend.background = element_rect(fill = "white", color = "white"),
        legend.key = element_rect(fill = "white", color = "white"),
        legend.position = "bottom",
        legend.box = "horizontal")

ggplotly(my_plot, tooltip = c("y", "fill"))
```

It can be seen that american ind are very less in quantity but the officers who has high experience belongs to this category.

Criminals

Now let see the count of racial category of criminals that which category has most criminals.

```
#Now let see the racial category of criminals
library(plotly)

race_count <- data.frame(table(data$SUBJECT_RACE))

my_plot <- plot_ly(race_count, x = ~Var1, y = ~Freq, type = "bar",
             marker = list(color = "#1f77b4", opacity = 0.7)) %>%
  layout(title = "Count of Criminals by Racial category",
         xaxis = list(title = "Race"),
         yaxis = list(title = "Count"))

# printing result
my_plot
```

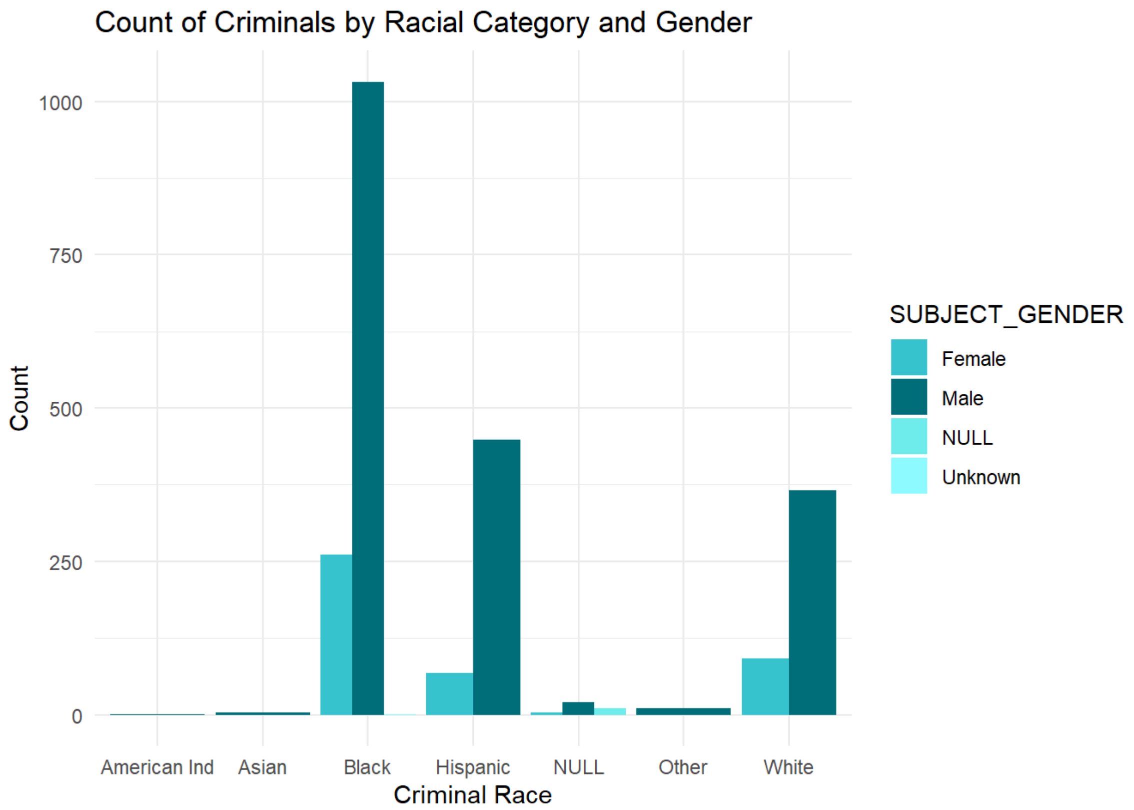
It can be seen that mostly Black people committed crimes and Hispanic and white people are having some way around same ratio of crime.

let find the genders of the criminals who committed crimes

```
#finding count of criminals by racial category and gender
race_gender_count <- data %>%
  group_by(SUBJECT_RACE, SUBJECT_GENDER) %>%
  summarize(n = n()) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'SUBJECT_RACE'. You can override using the
## `.groups` argument.
```

```
ggplot(race_gender_count, aes(x = SUBJECT_RACE, y = n, fill = SUBJECT_GENDER)) +
  geom_bar(stat="identity", position="dodge") +
  xlab("Criminal Race") +
  ylab("Count") +
  ggtitle("Count of Criminals by Racial Category and Gender") +
  scale_fill_manual(values = c("#38C4D0", "#026F78","#6FEBEF", "#8FFCFF")) +
  theme_minimal()
```



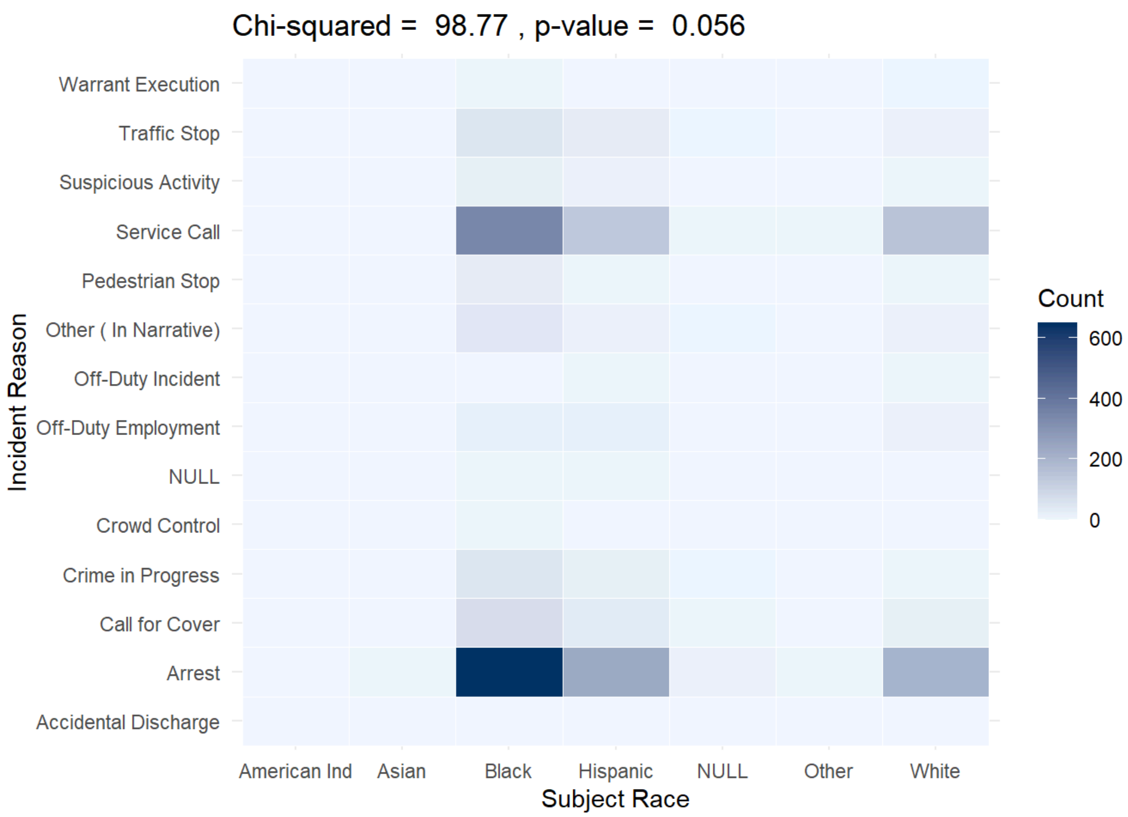
The male genders commited more crimes where as maximum womens also belongs to black racial category who commited more crimes.

lets do correlation analysis between Incident Reason and Criminal Race

```
# Create a contingency table between INCIDENT_REASON and SUBJECT_RACE
table <- table(data$INCIDENT_REASON, data$SUBJECT_RACE)

# Calculate the chi-squared test statistic and p-value
chi_sq <- chisq.test(table)
```

```
## Warning in chisq.test(table): Chi-squared approximation may be incorrect
```

```
melted_table <- melt(table)

# Create a heatmap of contingency table
ggplot(melted_table, aes(x = Var2, y = Var1, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "#F0F8FF", high = "#003366") +
  theme_minimal() +
  labs(x = "Subject Race", y = "Incident Reason", fill = "Count") +
  ggtitle(paste("Chi-squared = ", round(chi_sq$statistic, 2), ", p-value = ", for
mat.pval(chi_sq$p.value, digits = 2)))
```



According to a chi-square value of 98.77 and a p-value of 0.056.There is some evidence of a relationship between the two variables being examined, but not enough to reject the null hypothesis.

lets find how many criminals and officers are injured during this year 2016

```
# checking the comparision of criminal and officer injuries
injury_count <- data.frame(
  type = c("Criminal", "Officer"),
  yes_count = c(sum(data$SUBJECT_INJURY == "Yes"), sum(data$OFFICER_INJURY == "Ye
s"))
)

# maing bar chart
plot_ly(injury_count, x = ~type, y = ~yes_count, type = "bar",
        marker = list(color = c("#38C4D0", "#026F78"))) %>%
  layout(title = "Comparison of Criminal and Officer Injuries",
         xaxis = list(title = "Injury Type"),
         yaxis = list(title = "Count of 'Yes' Injuries"))
```

It can be seen that criminals were getting more injured compare to police officers. 620 criminals and 229 officers were injured.

IF there are more criminals then it is necessary to find what is the most frequent reason to use the force against criminal

```
#finding the reason why force used
reason_count <- data.frame(table(data$REASON_FOR_FORCE))

my_plot <- plot_ly(reason_count, x = ~Var1, y = ~Freq, type = "bar") %>%
  layout(title = "Count of Reasons for Use of Force",
         xaxis = list(title = "Reason for Force"),
         yaxis = list(title = "Count"))

# printing the plot
my_plot
```

The mostly cases are of arrest for which force needed to be used where as active aggression and danger to self or others are quit same.

lets find on monthly basis how many criminals were get arrested by force

```
#creating a line graph to see how many criminal arrested per month
df_count <- data %>%
  filter(SUBJECT_WAS_ARRESTED == "Yes") %>%
  group_by(month) %>%
  summarise(count = n())

plot_ly(df_count, x = ~month, y = ~count, type = "scatter", mode = "lines+marker
s") %>%
  layout(title = "Count of Arrested criminals per Month", xaxis = list(title = "M
onth"), yaxis = list(title = "Count of Arrested"))
```

it is clearly shown that initially the arresting trend was on its peak but the number of arresting was decreasing rapidly after september.

Now lets analyze which Division has reported highest crime

```
# Creating a pie chart to see the count of crimes in DIVISION
gender_count <- data.frame(table(data$DIVISION))

# Create an interactive pie chart using plotly
my_plot <- plot_ly(gender_count, labels = ~Var1, values = ~Freq, type = "pie",
             marker = list(colors = c("#38C4D0", "#026F78"), line = list(color =
"#FFFFFF", width = 1))) %>%
  layout(title = "Count of Crimes in Division in a year")

my_plot
```

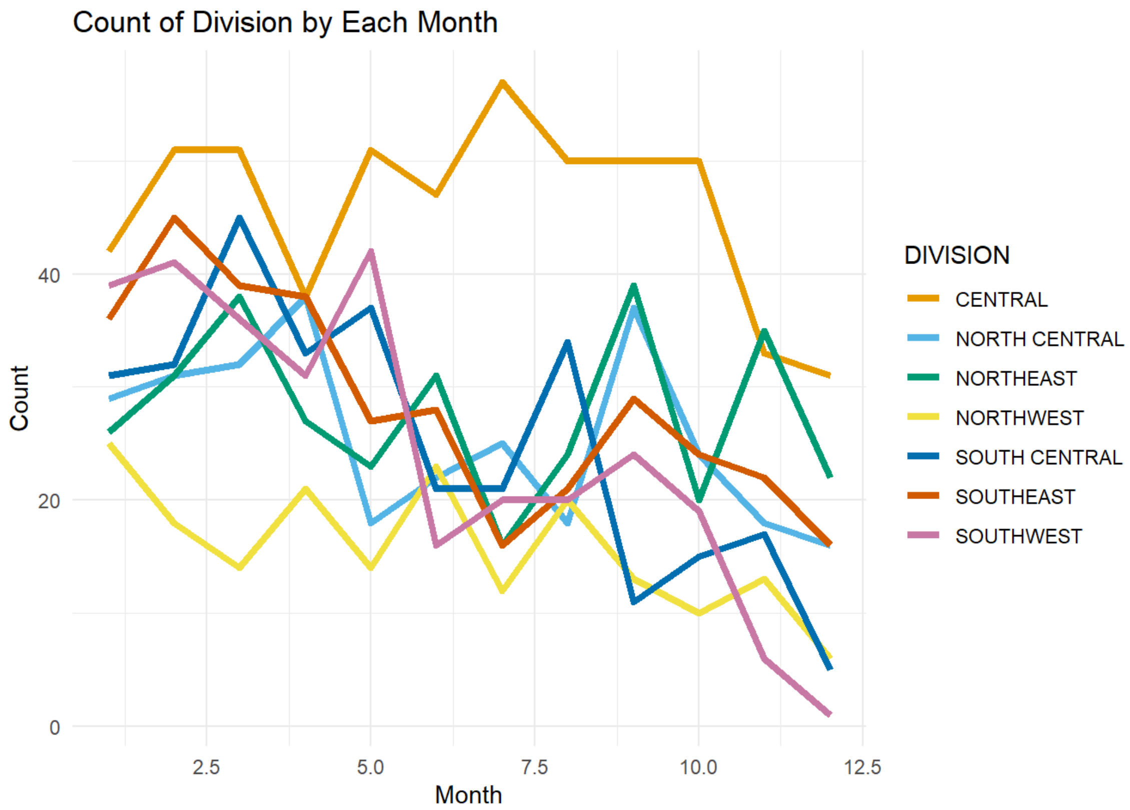
It can be seen that the central division reported the highest crime where as northwest has the lowest crime reported.Let se the trend of crimes in division in every month.

```
# finding the count of division respect to each month
division_month_count <- data %>%
  group_by(DIVISION, month) %>%
  summarize(count = n())
```

```
## `summarise()` has grouped output by 'DIVISION'. You can override using the
## `.groups` argument.
```

```
# Create the line graph
ggplot(division_month_count, aes(x = month, y = count, color = DIVISION)) +
  geom_line(size = 1.5) +
  scale_color_manual(values = c("#E69F00", "#56B4E9", "#009E73", "#F0E442", "#007
2B2", "#D55E00", "#CC79A7")) +
  labs(title = "Count of Division by Each Month", x = "Month", y = "Count") +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



It can be seen that the central division has the highest crime rate from the beginning but we can see that by the time crime rate was coming down.

lets figure out the location district with each division.

```
# Group the data by "DIVISION", "LOCATION_DISTRICT", and "month" and calculate th
e count
division_district_month_count <- data %>%
  group_by(DIVISION, LOCATION_DISTRICT, month) %>%
  summarize(count = n())
```

```
## `summarise()` has grouped output by 'DIVISION', 'LOCATION_DISTRICT'. You can
## override using the `.groups` argument.
```

```
# Create the plotly plot
plot_ly(division_district_month_count, x = ~month, y = ~count, color = ~DIVISION,
        split = ~LOCATION_DISTRICT, type = "scatter", mode = "lines+markers") %>%
  layout(title = "Count of Division in Location District by Month",
         xaxis = list(title = "Month"), yaxis = list(title = "Count"),
         legend = list(orientation = "h"))
```

it can be seen that all central division has the higher rate of crime report but specifically D14 experienced higher trend amongst all

lets find out how many years of experience is good to reduce the crime

```
# create a line graph to see the experience officer in each month they worked
my_plot <- ggplot(data, aes(x = month, y = OFFICER_YEARS_ON_FORCE, group = 1)) +
  geom_line(color = "#13979B", size = 2) +
  labs(title = "Experience Officers in Force by Month",
       x = "Month",
       y = "Officer Years on Force") +
  theme(plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
        axis.text = element_text(size = 14),
        axis.title = element_text(size = 16),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        panel.background = element_blank(),
        axis.line = element_line(colour = "black", size = 1))
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2
3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
my_plot <- ggplotly(my_plot)

my_plot
```

It can be seen that the officers who have average 5 years or less experience that means they are young and have enough energy or potential to handle the crime and city situation.

lets find what are the reasons why criminal are getting arrested.

```
# finding the count of reason of arrrest
gender_count <- data.frame(table(data$SUBJECT_DESCRIPTION))

my_plot <- plot_ly(gender_count, labels = ~Var1, values = ~Freq, type = "pie",
             marker = list(colors = c("#38C4D0", "#026F78"), line = list(color =
"#FFFFFF", width = 1))) %>%
  layout(title = "Count of Reasons")

# Display the plot
my_plot
```

The maximum crime reason was reported as the mental disability after that alcohol and after that the drugs these are the reason why they did crime. lets find the reason behind the incident

```
# finding each category in incident reason
reason_count <- data %>%
  group_by(month, INCIDENT_REASON) %>%
  summarise(count = n()) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'month'. You can override using the
## `.groups` argument.
```

```
my_plot <- plot_ly(reason_count, x = ~month, y = ~count, color = ~INCIDENT_REASO
N,
             type = "scatter", mode = "lines+markers") %>%
  layout(title = "Count of Incidents by Reason in a Year",
         xaxis = list(title = "Month"),
         yaxis = list(title = "Count"))

my_plot
```

```
## Warning in RColorBrewer::brewer.pal(N, "Set2"): n too large, allowed maximum f
or palette Set2 is 8
## Returning the palette you asked for with that many colors

## Warning in RColorBrewer::brewer.pal(N, "Set2"): n too large, allowed maximum f
or palette Set2 is 8
## Returning the palette you asked for with that many colors
```

we can see that the major reason behind the incident was arrest and service call

lets plot a scatter plot with longitude and lattitude to see the areas where incident happened.

```
# Create a scatter plot using ggplot2
my_plot <- ggplot(data, aes(LOCATION_LONGITUDE, LOCATION_LATITUDE,)) +
  geom_point(size = .75, alpha = 0.7, show.legend = FALSE) +
  scale_color_manual(values = color_palette) +
  labs(x = "Longitude", y = "Latitude", title = "Scatter plot of Incident Areas
") +
  theme_minimal(base_size = 14) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank())

# Convert the ggplot object to plotly and add interactivity
my_plot <- ggplotly(my_plot, tooltip = c("x", "y", "color")) %>%
  labs(title = "Scatter plot of Incident Areas",
       x = "Longitude",
       y = "Latitude",
       hoverlabel = list(bgcolor = "white", font = list(family = "Arial", size =
12)))

my_plot
```

```
## [[1]]
##
## $x
## [1] "Longitude"
##
## $y
## [1] "Latitude"
##
## $hoverlabel
## $hoverlabel$bgcolor
## [1] "white"
##
## $hoverlabel$font
## $hoverlabel$font$family
## [1] "Arial"
##
## $hoverlabel$font$size
## [1] 12
##
##
##
## $title
## [1] "Scatter plot of Incident Areas"
##
## attr(,"class")
## [1] "labels"
```

let use map to see to see these areas in a map view

Lets plot the map using the longitude and latitudes points to see where those incident happened.

```
# creating a map using longitude and lattitude
m <- leaflet() %>%
  addProviderTiles("Esri.WorldTopoMap") %>%
  setView(lng = data$LOCATION_LONGITUDE[1],
          lat = data$LOCATION_LATITUDE[1],
          zoom = 10)

# Add circles to the map for each point in the dataset
m %>% addCircleMarkers(data = data,
                       lng = ~LOCATION_LONGITUDE,
                       lat = ~LOCATION_LATITUDE,
                       radius = 5,
                       color = "#3182bd",
                       fillOpacity = 0.7,
                       fillColor = "#9ecae1",
                       popup = ~paste("Location:", LOCATION_LATITUDE, LOCATION_LONGITUDE))
```

Here we can see the areas which are highlighted

SUMMARY

The center of police Equity (CPE) used data science to build their system more fair. They analyze the data and locate those areas where racial discrimination exists and cannot be accounted by crime and poverty rates.

By examining crime data from November 2016 to March 2017, it is evident that crime rates began to decline after the month of November, and 2017 started off to a far better start than 2016. The maximum number of incident reported from 21st February to 6th March, with a peak of 144 cases.When police officers are analyzed by their demographics, it becomes clear that 89.8% of them are male and only 10.2% are female. American Indian and Asian police are the least represented, and white officers make up the majority of the force.When examined the dataset it is clear that black people are the largest group of offenders, followed by Hispanic and white people. The majority of offenders are male and black female.

With 620 criminals and 229 police officers suffering injuries in the same year, the data also demonstrates that criminals are more likely to suffer injuries than law enforcement officers.Arrest is the most prevalent justification for using force against criminals; active violence and danger to oneself or others are less frequent justifications. The data also demonstrates that, after September, the trend of forced arrests began to diminish.

The central division reported the most crimes, and among all the districts, D14 had the greatest tendency of crime. Unexpectedly, it was shown that cops with an average tenure of 5 years or less were more successful in lowering crime.Finally, it was discovered that mental illness, followed by alcohol and drug abuse, was the most frequent cause of criminal conduct. Future actions and initiatives to lower crime rates can be influenced by knowing these causes of criminal behavior.

Organizations like the CPE are aiming to establish more fair and just systems that bridge the gap between law enforcement and communities and eventually improve public safety, community trust, and racial justice by utilizing data and science to inform police policies.